

Data Structures & Algorithms Complete Roadmap

Phase 0: Programming Foundations

- Variables & Data Types
- Operators
- Input / Output
- Conditional Statements
- Loops
- Functions / Methods
- Introduction to Time & Space Complexity
- Number System & Basic Mathematics
- Logarithms, Powers, Modulus Arithmetic

Phase 1: Complexity Analysis

- What is an Algorithm?
- Time Complexity
- Space Complexity
- Big-O Notation
- Best, Average, Worst Case Analysis
- Common Complexities: $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, $O(2^n)$

Phase 2: Arrays & Strings

- 1D and 2D Arrays
- Prefix Sum Technique
- Sliding Window Technique
- Two Pointer Technique
- Kadane's Algorithm
- Subarrays & Subsequences
- String Operations
- Palindrome & Anagram Problems
- Frequency Counting
- Basic Pattern Matching

Phase 3: Searching & Sorting

- Linear Search

- Binary Search
- Binary Search on Answer
- Lower Bound & Upper Bound
- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Counting & Radix Sort

Phase 4: Recursion & Backtracking

- Recursion Basics & Call Stack
- Base Conditions
- Recursive Tree
- Tail Recursion
- Backtracking Concept
- Subsets & Permutations
- Combination Sum
- N-Queens
- Sudoku Solver

Phase 5: Linked List

- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Insertion & Deletion
- Reverse Linked List
- Middle of Linked List
- Detect Cycle (Floyd's Algorithm)
- Merge Two Sorted Lists

Phase 6: Stack & Queue

- Stack Implementation
- Infix to Postfix/Prefix
- Valid Parentheses
- Next Greater Element

- Stock Span Problem
- Queue Implementation
- Circular Queue
- Deque
- Priority Queue

Phase 7: Hashing

- HashMap & HashSet
- Frequency Map
- Collision Handling
- Open Addressing vs Chaining
- Two Sum Problem
- Longest Consecutive Sequence
- Subarray Sum Equals K
- Majority Element

Phase 8: Trees

- Tree Terminology
- Binary Tree
- Binary Search Tree
- Tree Traversals (Inorder, Preorder, Postorder)
- Level Order Traversal
- Height & Diameter
- Lowest Common Ancestor
- Validate BST

Phase 9: Heaps & Priority Queue

- Min Heap & Max Heap
- Heapify
- Heap Sort
- Kth Largest/Smallest Element
- Merge K Sorted Arrays

Phase 10: Graphs

- Graph Representation
- Directed & Undirected Graphs

- BFS & DFS
- Cycle Detection
- Topological Sort
- Dijkstra's Algorithm
- Bellman-Ford Algorithm
- Minimum Spanning Tree
- Disjoint Set (Union Find)

Phase 11: Greedy Algorithms

- Activity Selection
- Job Scheduling
- Fractional Knapsack
- Huffman Coding
- Minimum Platforms Problem

Phase 12: Dynamic Programming

- DP Concepts
- Memoization
- Tabulation
- 0/1 Knapsack
- Coin Change
- Longest Common Subsequence
- Longest Increasing Subsequence
- Matrix Chain Multiplication
- DP on Grids & Trees

Phase 13: Advanced Topics

- Bit Manipulation
- Trie Data Structure
- Segment Tree
- Fenwick Tree (BIT)
- Advanced Sliding Window
- Mo's Algorithm

Phase 14: Interview Preparation

- Pattern-based Problem Solving

- Mock Interviews
- Clean Code Practices
- Explaining Approach Clearly
- Handling Edge Cases